Transformer condition report - DT-001

As of 23 Apr 2026 15:00, covering 01 Apr 2026 00:00 to 23 Apr 2026 15:00 (22.6 days). Generated 08 Oct 2026 16:01.

Nameplate		Data source	
Rating	100 kVA (sidecar showcase.mapping.json)	File	showcase.csv
Voltage	11 kV / 433 V, ONAN	Rows	read 2880, kept 2880
Rated current	133.3 A	Cadence	15.0 min
Top-oil / hot-spot rise	55 K / 25 K	Absent channels	none
Oil time constant	3 h	Thermal diagnosis	available

Summary

Status: ALARM Health index 72/100 Active alerts 1

Cooling path degraded · rise +17% over design · inspect radiators, oil level and fans

Health component	Value	Score	100 at	0 at (alarm)	Weight
Thermal ageing (F_AA)	0.074 pu	100	<= 0.125	1	0.30
Cooling (rise-fit slope)	1.168	0	<= 1.025	1.06	0.25
Loading (K)	0.508 pu	100	<= 0.8	1	0.20
Measurement integrity (1 h)	97.500 %	80	<= 100	87.5	0.15
Phase deviation	0.741 %	100	<= 1.45	4	0.10

Projection at present load 51% and ambient 44.7 C (cooling factor 1.168): hot-spot settles at 83.2 C, stays under 110 C. Existing 60 C oil trip, for comparison: already exceeded.

Recommendations

P1 Inspect radiators for fouling, check oil level and fans. Schedule within days, not weeks. (Cooling path degraded)

Insulation life consumed

Window	Aged hours	Real hours	Rate (x rated)	Value (Rs)
Last hour	0.10	1.2	0.078	0
Last 24 h	0.61	24.2	0.025	2
Whole period	84.56	543.2	0.156	211

IEEE C57.91 ageing acceleration, thermally upgraded paper, reference hot-spot 110 C. Rs 450,000 replacement cost / 180,000 h rated life = Rs 2.50 per aged hour.

Fault log

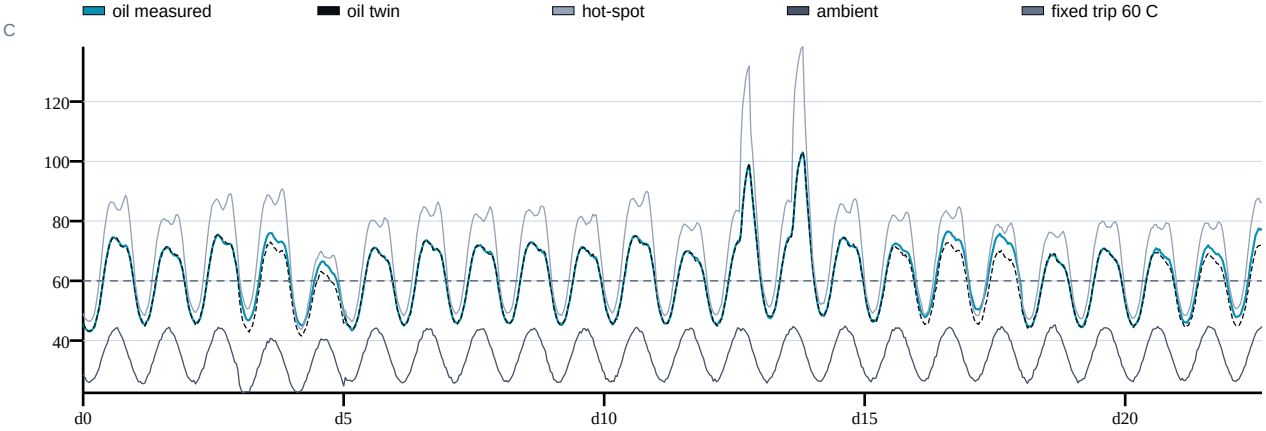
Event	Family	Start	End	Diagnosis	Action
Thermal model mismatch	other	04 Apr 04:00	04 Apr 22:45	Measured rise is lower than the nameplate predicts at this load.	Check the nameplate rating and the oil probe placement.
Temperature instrument offset (oil or ambient probe)	sensor	05 Apr 01:00	06 Apr 00:45	Measured rise is shifted by +3.1 K at every load level; a real thermal fault would scale with load instead.	Recalibrate or replace the probe. Do not dispatch for a transformer fault; treat oil-temperature trips as unreliable until fixed.
Phase load unbalance growing	asset	08 Apr 21:45	08 Apr 22:30	Phase currents have moved apart (peak 3.8% unbalance) and the voltages confirm a real redistribution of load, not a CT error.	Rebalance single-phase loads across R/Y/B at the next feeder visit.
Phase load unbalance growing	asset	09 Apr 01:00	09 Apr 02:00	Phase currents have moved apart (peak 4.1% unbalance) and the voltages confirm a real redistribution of load, not a CT error.	Rebalance single-phase loads across R/Y/B at the next feeder visit.
IB disagrees with the other phases	sensor	09 Apr 07:30	09 Apr 08:15	Cross-checked against the other channels; the transformer state is unaffected.	Check the CT/VT ratio and connections on this phase.
Phase load unbalance growing	asset	09 Apr 08:30	09 Apr 11:15	Phase currents have moved apart (peak 5.3% unbalance) and the voltages confirm a real redistribution of load, not a CT error.	Rebalance single-phase loads across R/Y/B at the next feeder visit.

Event	Family	Start	End	Diagnosis	Action
Phase load unbalance growing	asset	09 Apr 11:45	12 Apr 00:15	Phase currents have moved apart (peak 10.3% unbalance) and the voltages confirm a real redistribution of load, not a CT error.	Rebalance single-phase loads across R/Y/B at the next feeder visit.
Overload	asset	13 Apr 15:00	13 Apr 19:00	Load reached 119% of rating.	Shed or transfer load; see the time-to-trip countdown.
Overload	asset	14 Apr 15:00	14 Apr 20:00	Load reached 123% of rating.	Shed or transfer load; see the time-to-trip countdown.
Temperature instrument offset (oil or ambient probe)	sensor	17 Apr 03:00	19 Apr 00:45	Measured rise is shifted by +2.1 K at every load level; a real thermal fault would scale with load instead.	Recalibrate or replace the probe. Do not dispatch for a transformer fault; treat oil-temperature trips as unreliable until fixed.
Cooling path degraded	asset	22 Apr 04:00	22 Apr 06:45	Oil rise is 7% above what the load and ambient explain (fit slope 1.07); the measured rise scales with load, so the instrument is not the cause.	Inspect radiators for fouling, check oil level and fans. Schedule within days, not weeks.
Cooling path degraded	asset	22 Apr 11:00	active	Oil rise is 8% above what the load and ambient explain (fit slope 1.08); the measured rise scales with load, so the instrument is not the cause.	Inspect radiators for fouling, check oil level and fans. Schedule within days, not weeks.
IY frozen at a constant value	sensor	23 Apr 09:15	23 Apr 14:00	Cross-checked against the other channels; the transformer state is unaffected.	Reset the logger channel; values during this period were replaced.

13 episodes. The fixed 60 C oil-trip comparison is not an event and is not listed.

Thermal trend

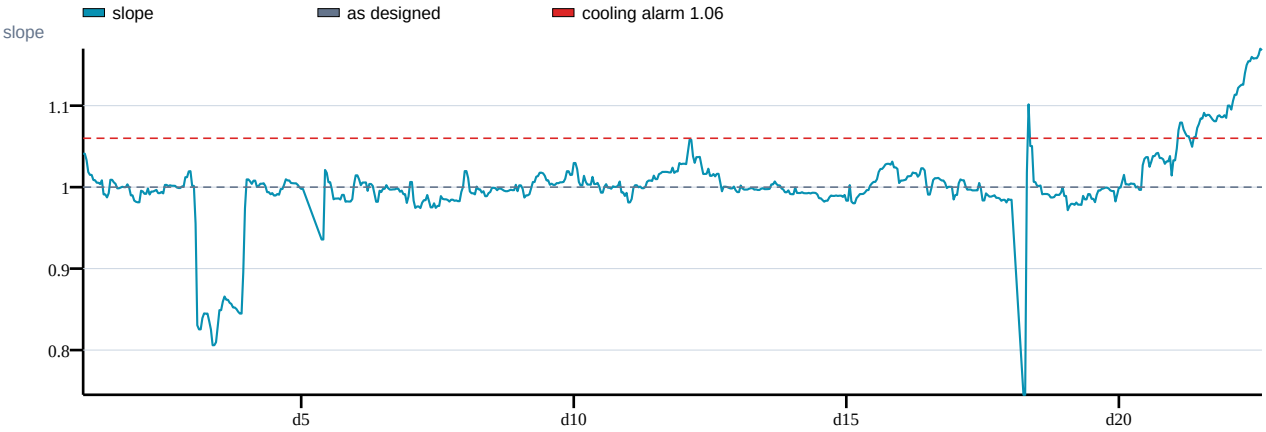
Top oil measured vs twin prediction, hot-spot and ambient (C)



Measured top oil should track the twin's prediction. A widening gap with load is a cooling problem; a constant gap is an instrument offset. Days since the start of the record on the horizontal axis.

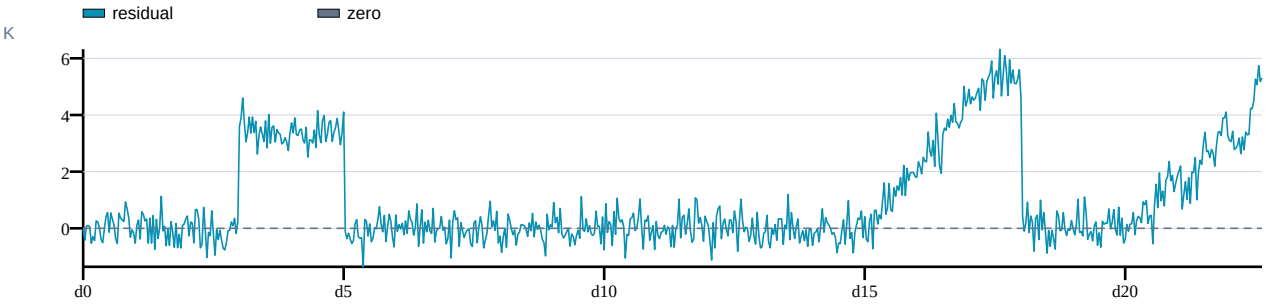
Cooling condition

Rise-fit slope: measured oil rise / predicted oil rise (1.00 = as designed)



Above 1.06 for three consecutive hourly fits is the cooling alarm. Brief spikes that coincide with a temperature step are the 24 h fit window straddling an instrument change (see the fault log), not a change in the transformer.

Thermal residual: measured - twin top oil (K)



Protection-setting comparison

Steady load each protection philosophy permits. A fixed 60 C oil trip carries the ambient inside it; a trip on rise over measured ambient (30 K, the protection the 60 C setting gives at 30 C) or on hot-spot (110 C) does not.

Ambient	Fixed 60 C oil	Rise 30 K	Hot-spot 110 C
15 C	0.86 pu	0.60 pu	1.12 pu
20 C	0.78 pu	0.60 pu	1.08 pu
25 C	0.69 pu	0.60 pu	1.04 pu
30 C	0.60 pu	0.60 pu	0.99 pu
35 C	0.50 pu	0.60 pu	0.95 pu
40 C	0.37 pu	0.60 pu	0.90 pu
42 C	0.31 pu	0.60 pu	0.88 pu
45 C (now)	0.19 pu	0.60 pu	0.86 pu
48 C	0.00 pu	0.60 pu	0.83 pu

Appendix - ingestion mapping report

Source showcase.csv. Separator ',', decimal '.', encoding utf-8-sig, header on line 1. Timestamp: timestamp -> text, month-first.

Channel	Source column	Matched by	Conf.	Unit	Factor	Unit reason
IR	IR	name	high	A	1	magnitude
IY	IY	name	high	A	1	magnitude
IB	IB	name	high	A	1	magnitude
VR	VR	name	high	V	1	magnitude
VY	VY	name	high	V	1	magnitude
VB	VB	name	high	V	1	magnitude
oil_temp	oil_temp	name	high	C	1	magnitude
ambient_temp	ambient_temp	name	high	C	1	magnitude

Rows read 2880, kept 2880; dropped: none.

Out-of-order rows sorted: 0.

Cadence 15.0 min median, regular.

Short holes interpolated (<= 2 samples): none.

Columns ignored: transformer_id.

Correct any wrong guess with python -m dtwin.ingest FILE --map CHANNEL=COLUMN --unit CHANNEL=UNIT --dayfirst yes|no --kva RATING --save-sidecar.